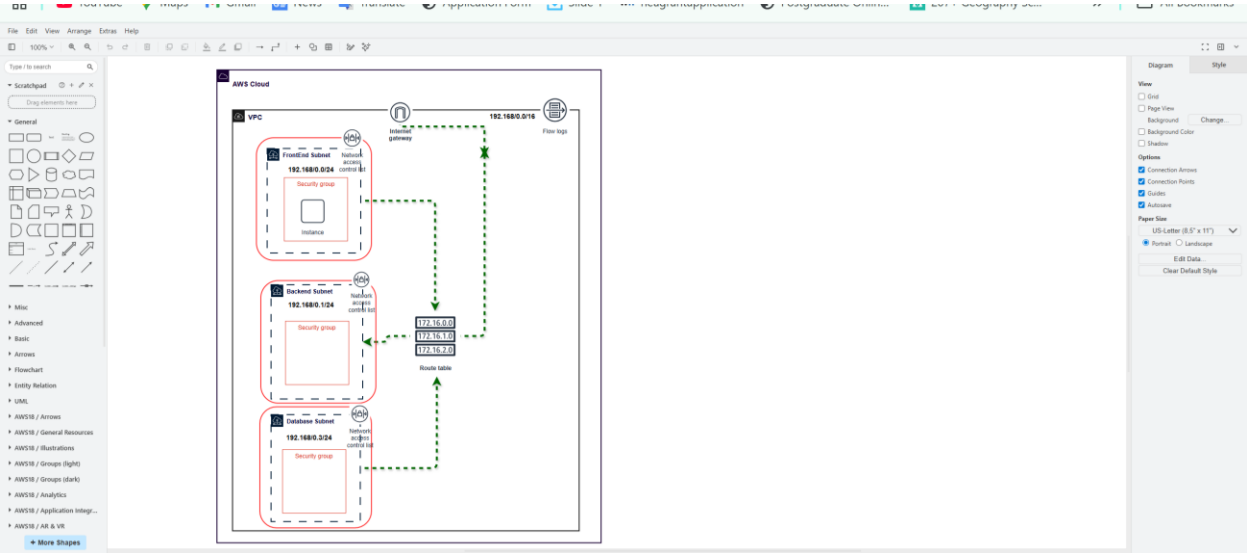
HOW TO CREATE A VPC IN AWS

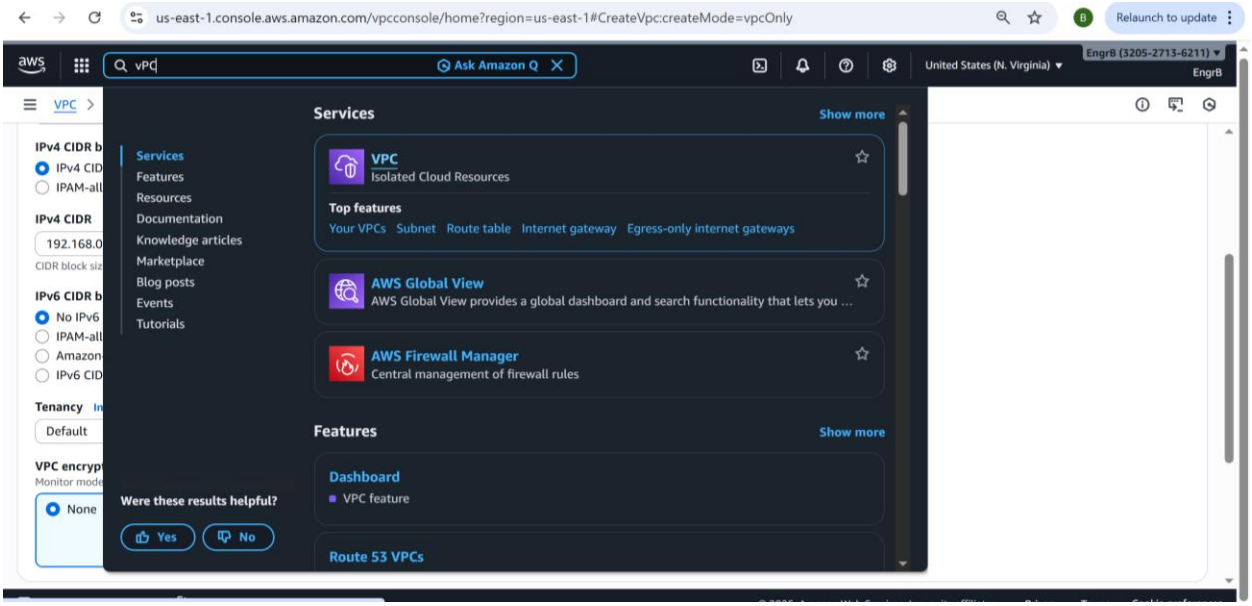
Introduction

This document provides a simple step-by-step guide on how to create a Virtual Private Cloud (VPC) in Amazon Web Services. A VPC is a logically isolated network that allows you to launch AWS resources in a virtual network that you define. When you create a VPC, you gain full control over your networking environment, including IP address ranges, subnets, route tables, and security settings. This documentation is intended for beginners and intermediate users who want to understand the foundational networking setup required to securely deploy and manage cloud-based applications within AWS.

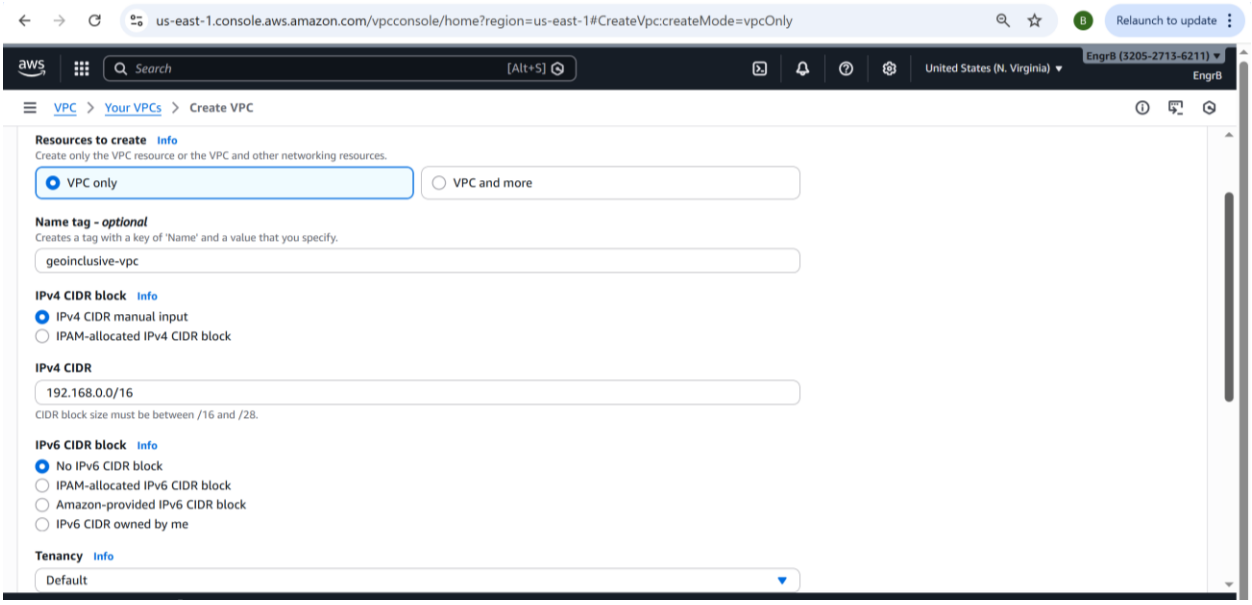
Step 1: Design your cloud Architecture using draw.io



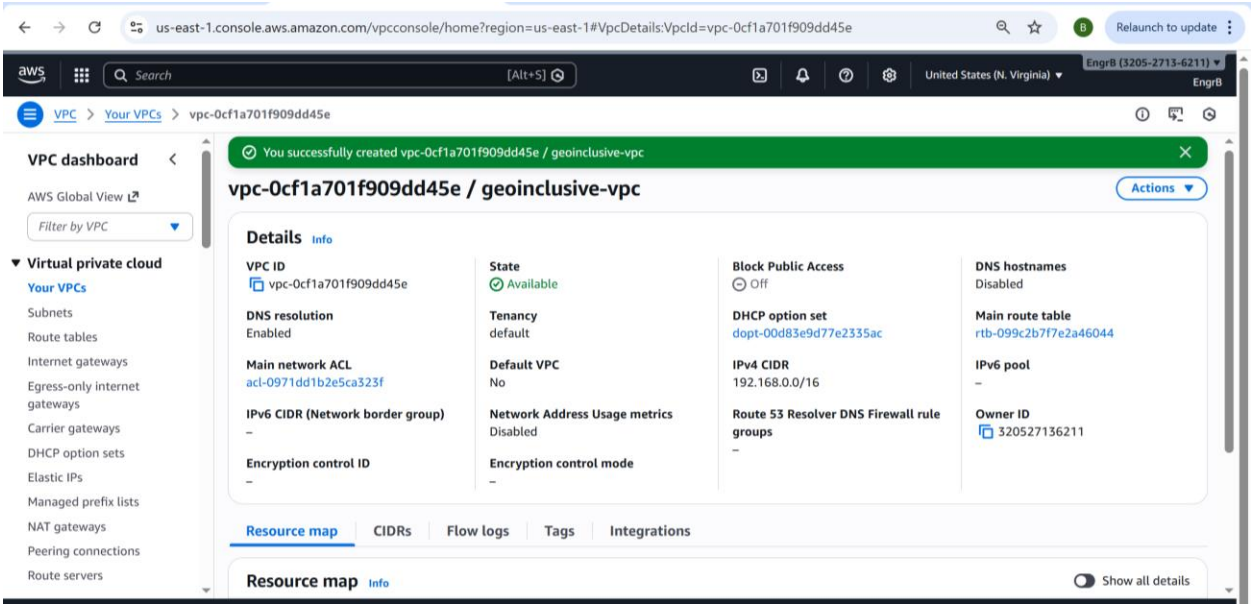
Step 2: After the design, log into your AWS profile, then using the search bar, search for VPC, and then click on it, then click on create VPC.



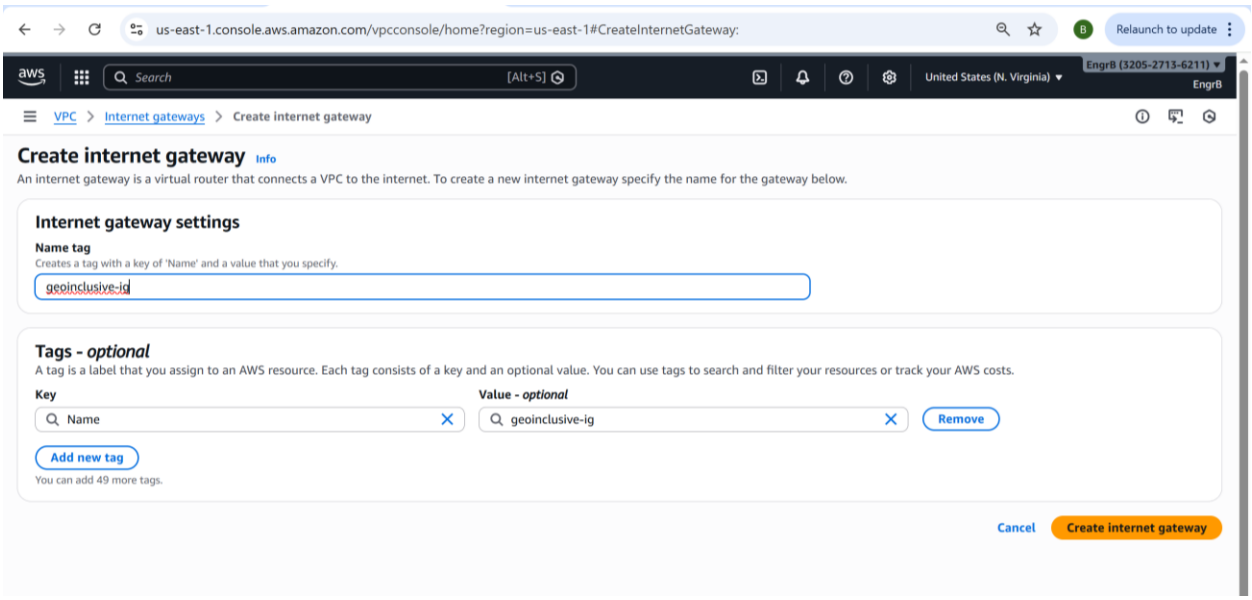
Step 3: When it opens, follow the instructions. Select a name for your VPC, input your IP Address, select default tenancy, and select none for VPC encryption control then create your VPC.



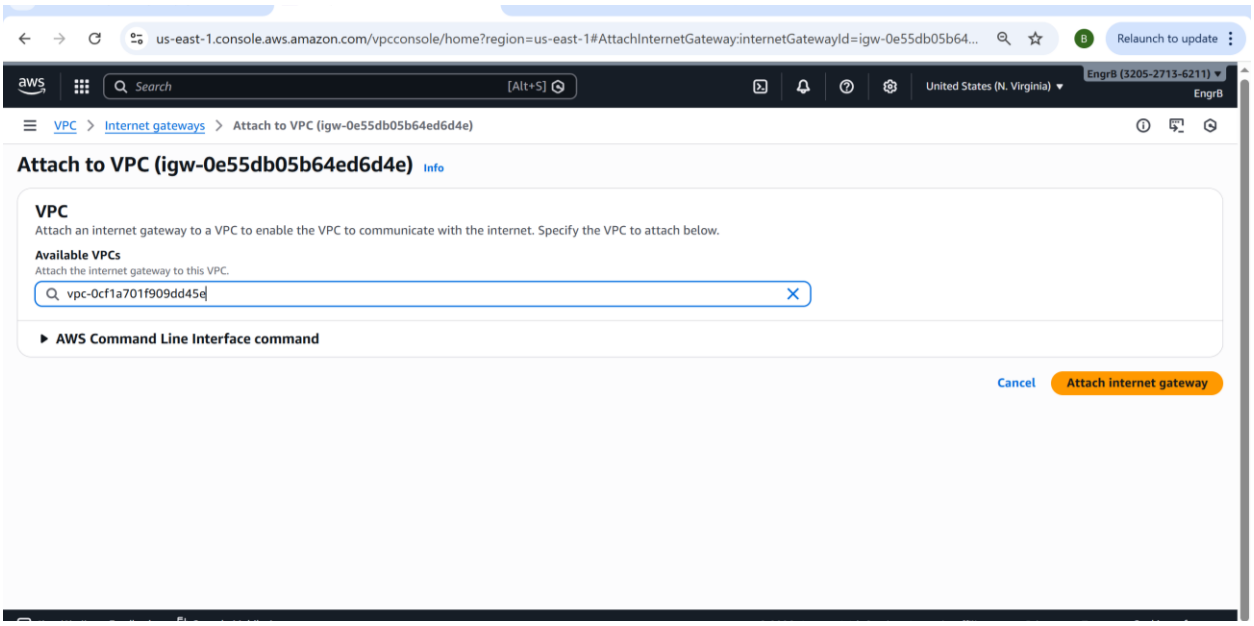
Step 4: The VPC will be created.



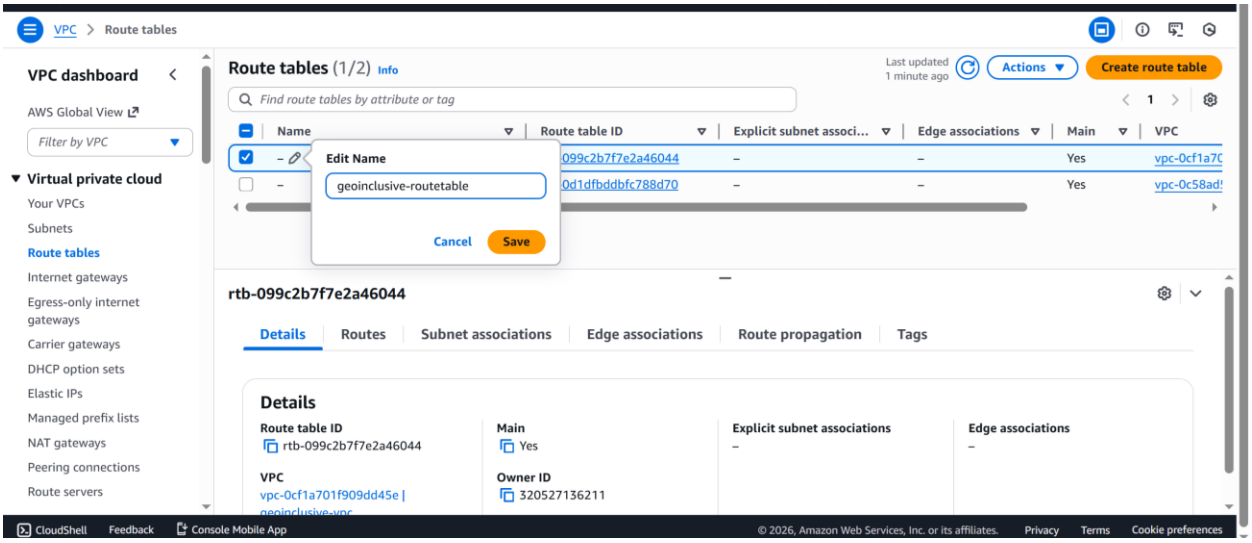
Step 5: Next, create an internet gateway by searching for it in the search bar. Click on create Internet gateway and name it, and then create



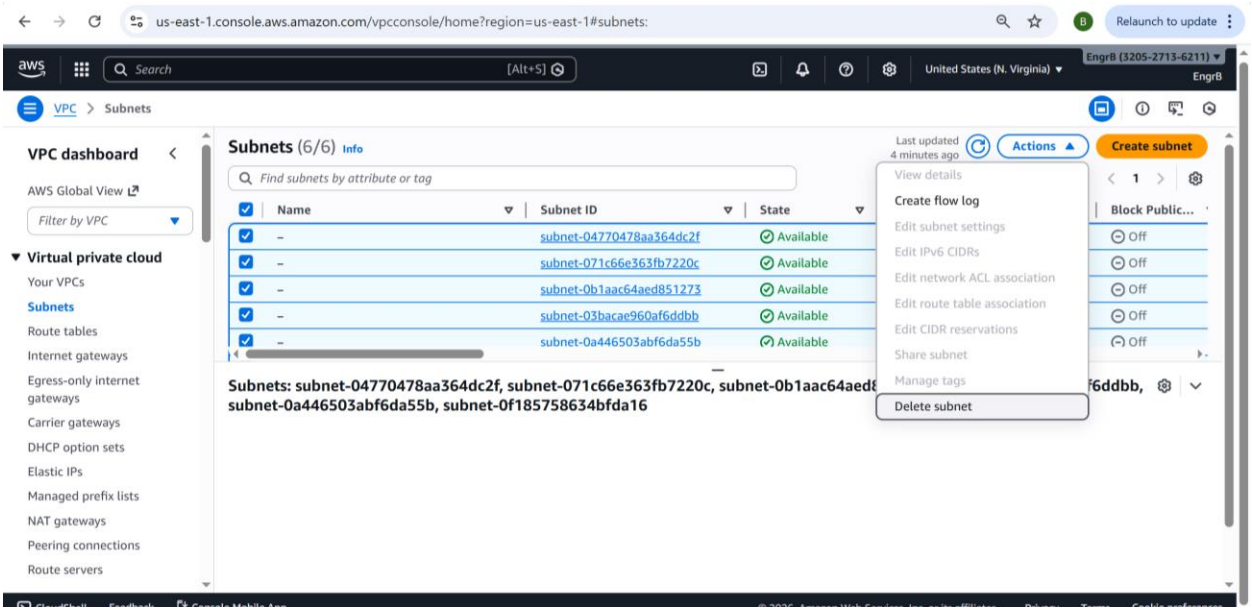
Step 6: After it is created, to attach it to the VPC, click on it, go to actions and click on attach to VPC.



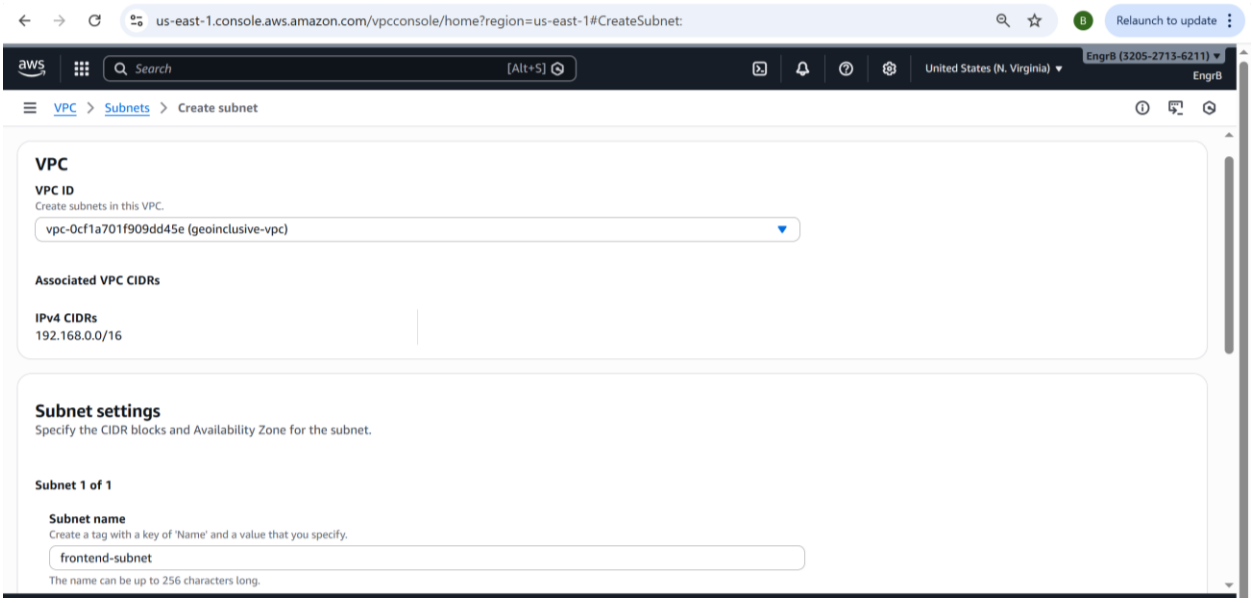
Step 7: Check your route table, and then edit the name to avoid confusion.



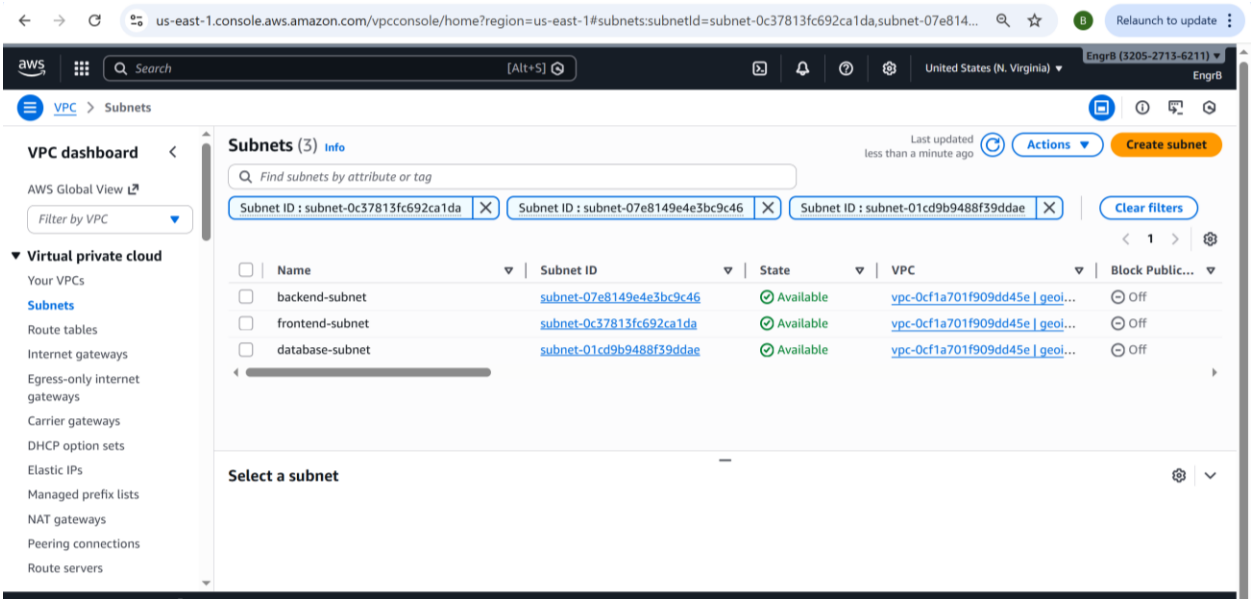
Step 8: After the attachment, click on subnets and then delete all the default subnets.



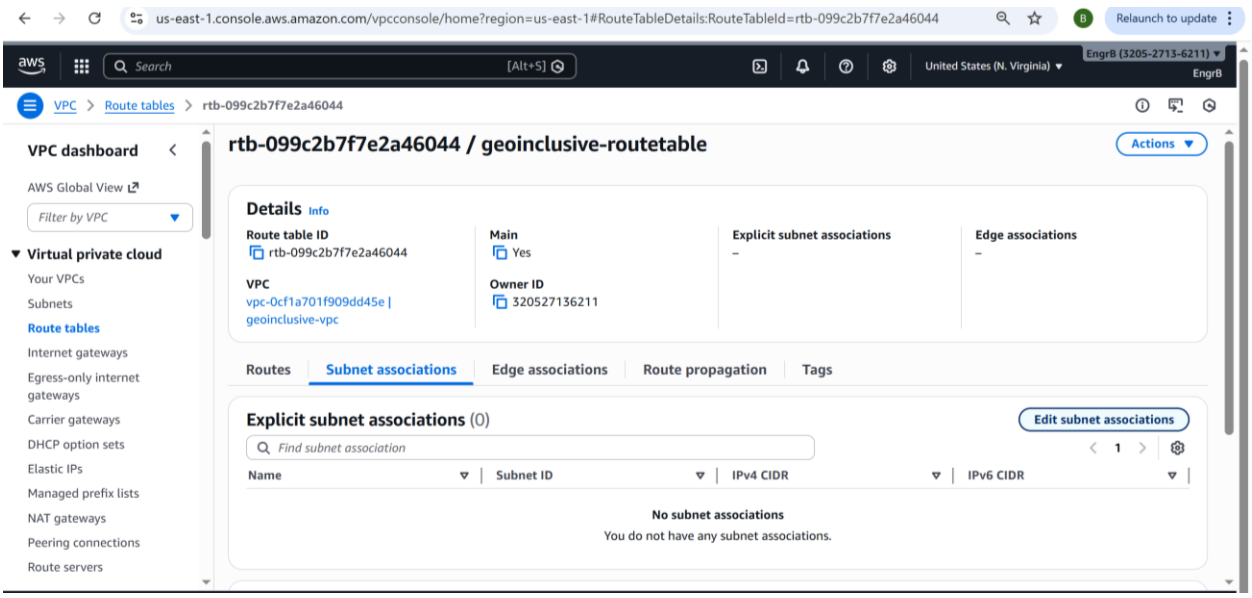
Step 9: Click on create subnet and fill in the information. Choose your VPC, add your name, select your zone, and create subnet.



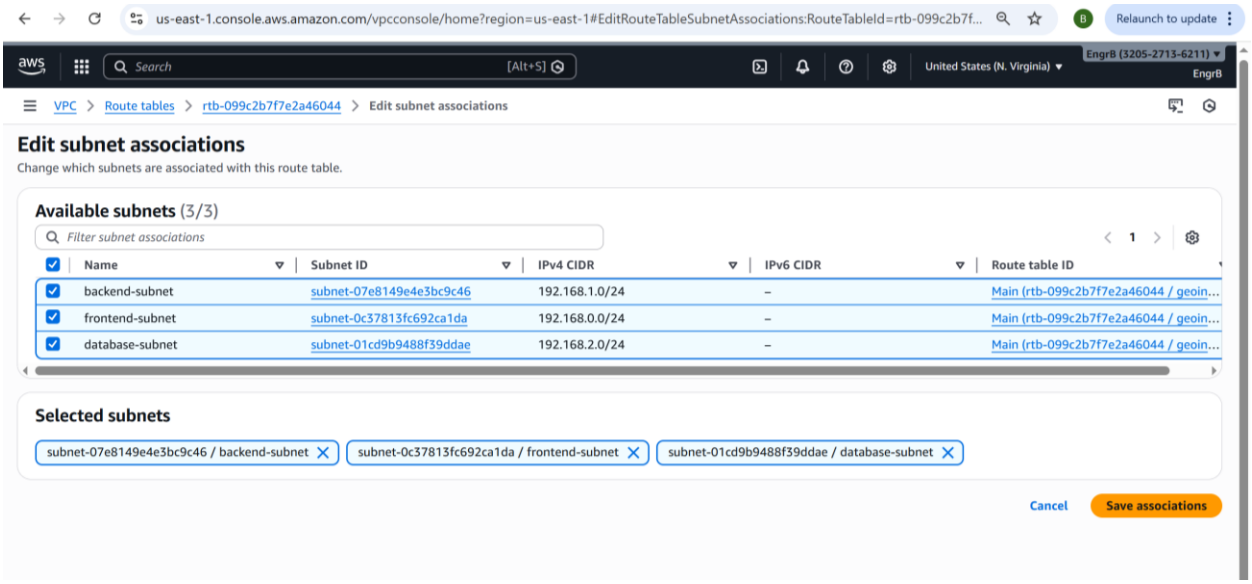
Step 10: You can click on add subnet, but ensure you chose another region different from the first subnet region. Create all Subnets and confirm them. You have successfully created your subnets

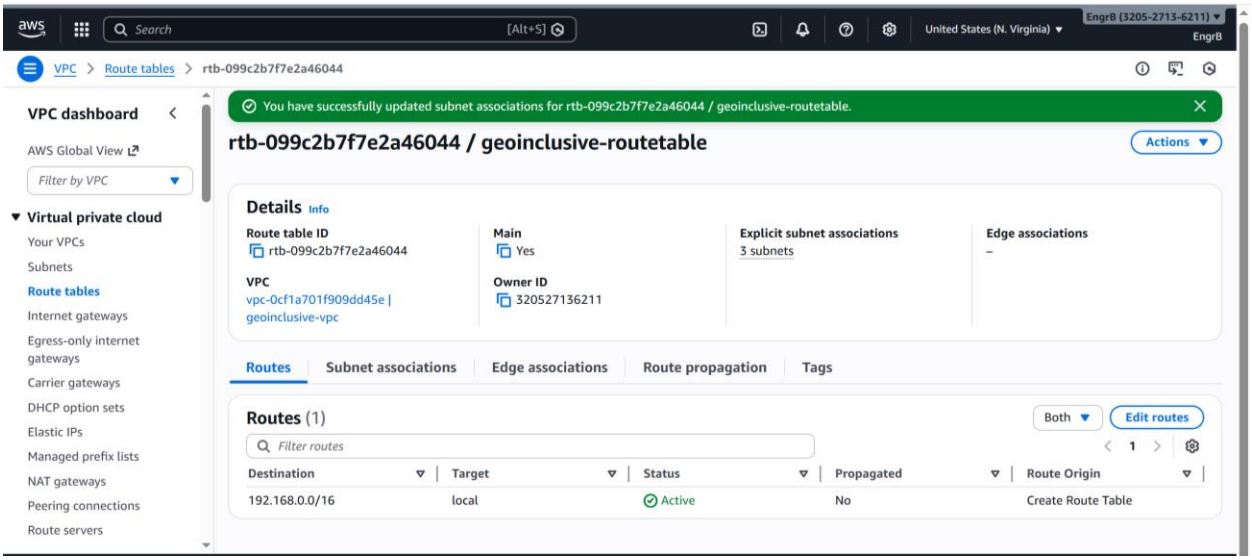


Step 11: Next is to attach the subnets to the route table. Go to the route table, click on it and check the subnet association, then click on edit associations.

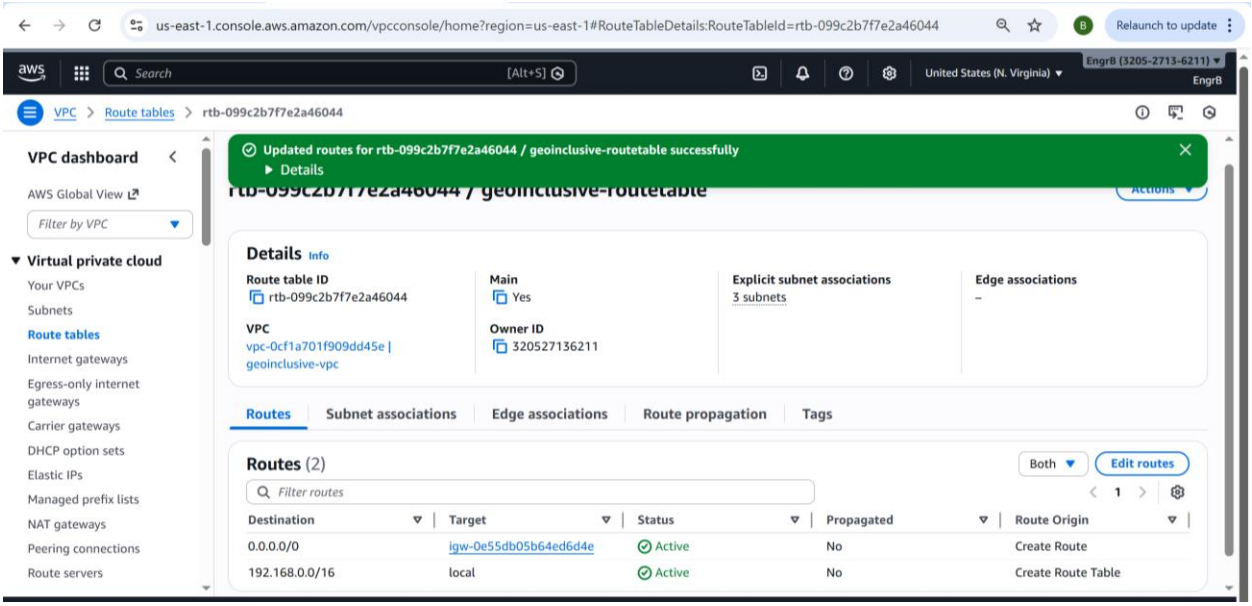


Step 13: Highlight all the subnets and click on save associations. Check to confirm.



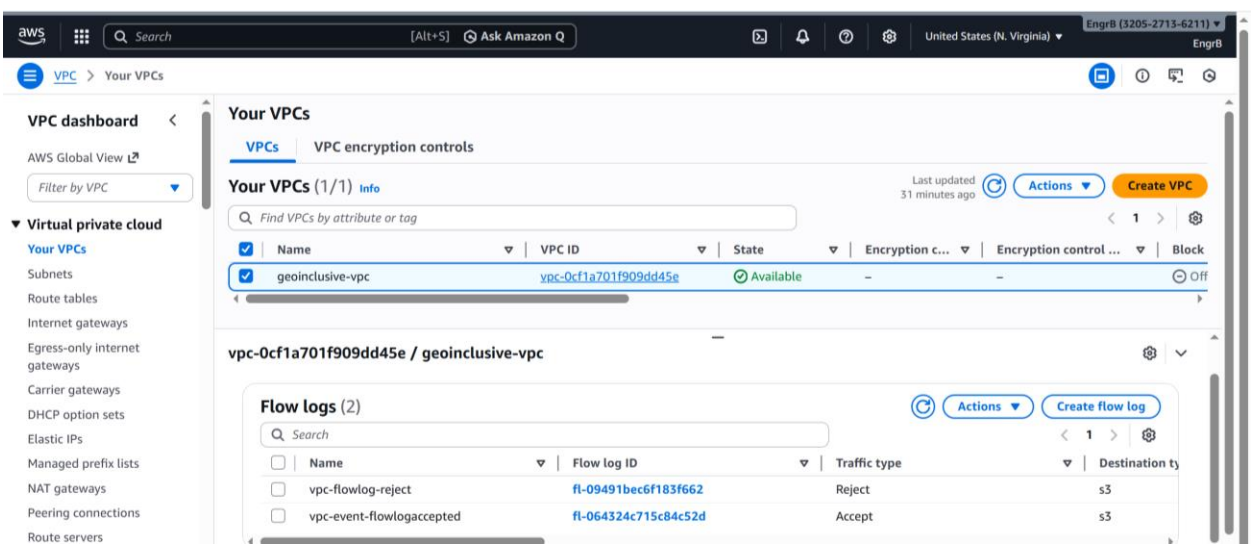


Step 14: Next is to create the Internet gateway to enable them access the internet. Go to the route table and check on route. Click on edit route and add a route. Select Internet gateway and then select the vpc-ig. Click on save.

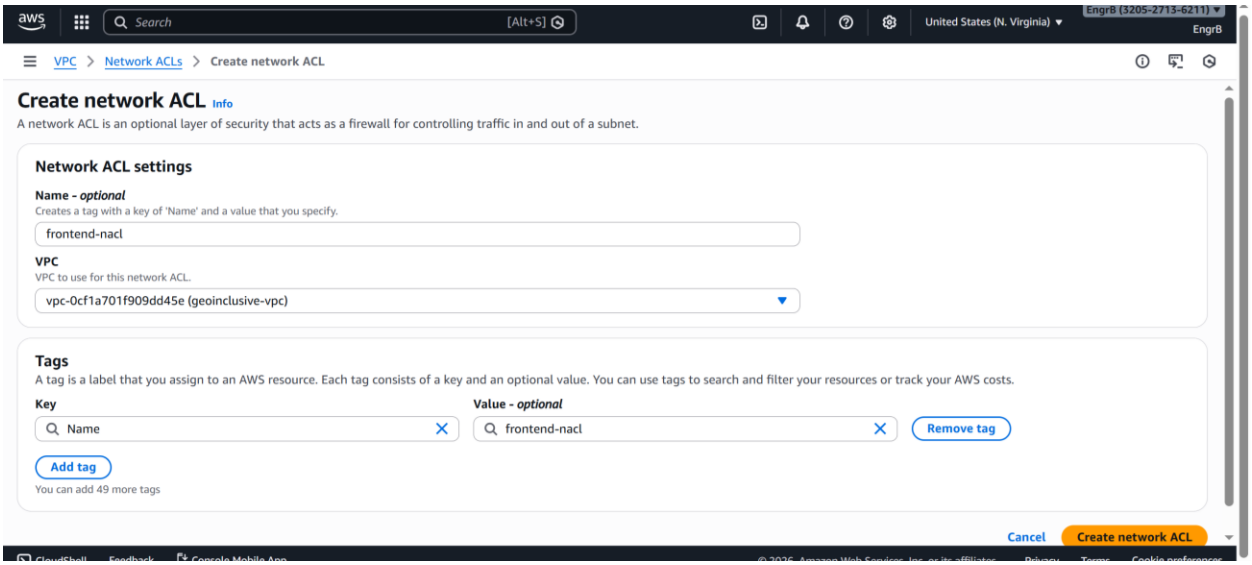


Step 15: Next, is to create the flowlog that will help to capture all traffic. Go to your VPC, open it, and go to flowlog, create a flowlog, give it a name, you can create the flowlog for accept and deny, choose the time, and you can send the data to any of your choice. (e.g Amazon s3 bucket). You can create your s3 bucket, open it, copy the Amazon resource name from the meta data and then paste it in the flow log and then create.

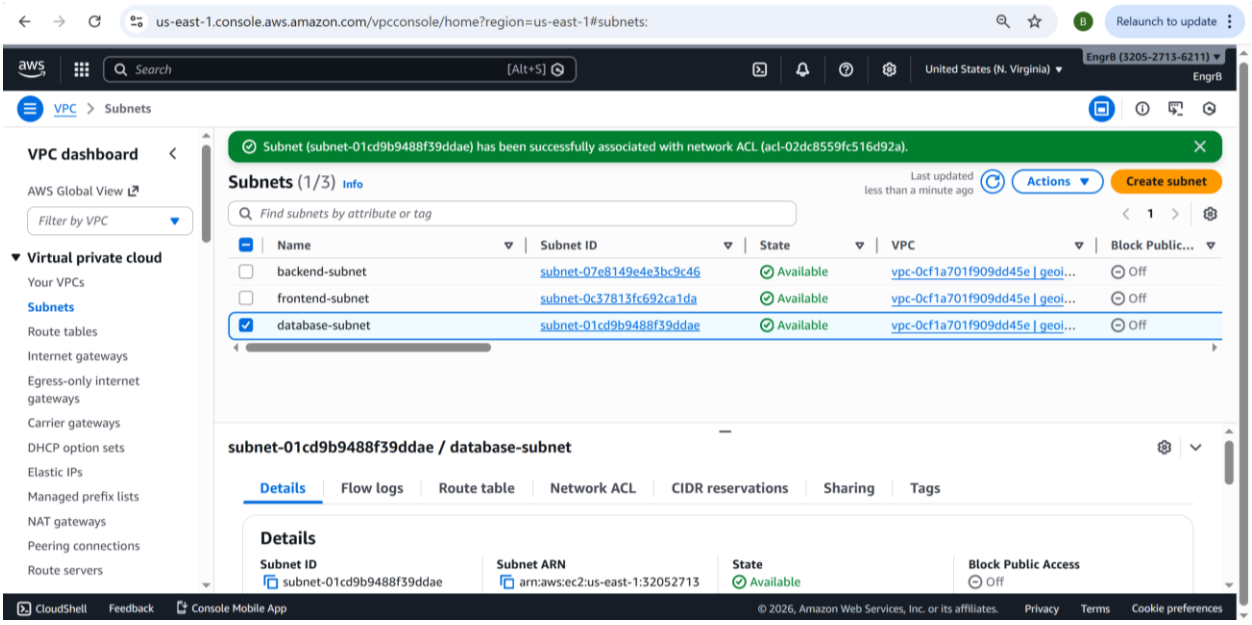
Then create another one for reject. Both will be captured inside your VPC.



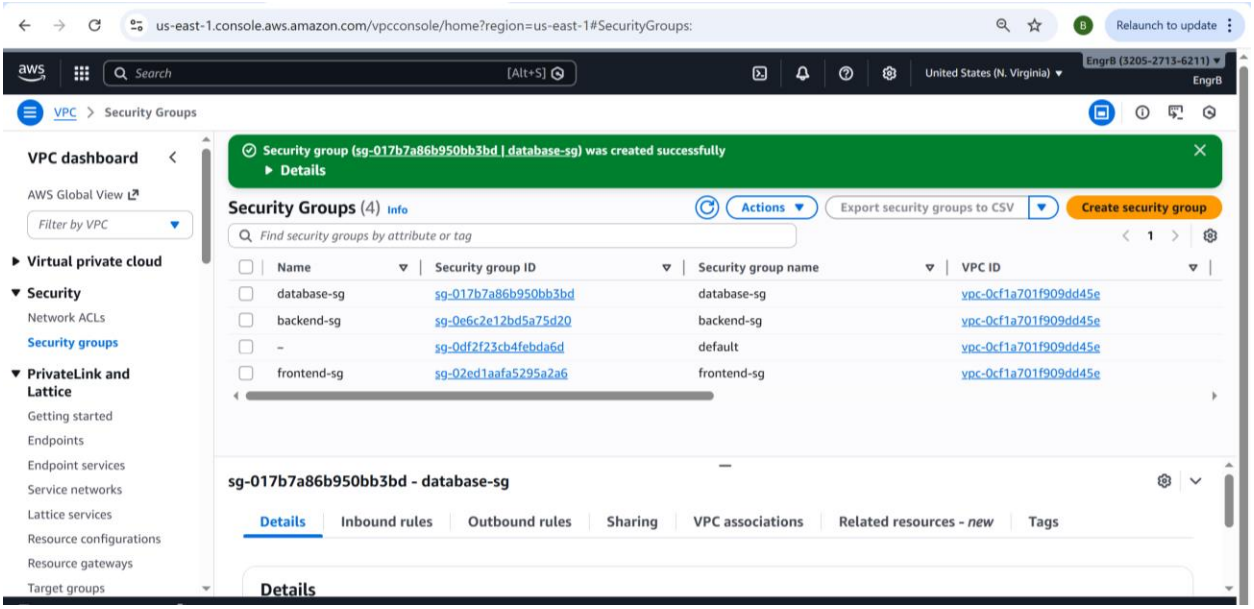
Step 16: Next is to create a NACL. For security reason, you need to put each subnet in their own NACLs. To do this, open the NACLs, click on create NACL, start with the front end and name it. And create. Repeat the same process for the rest subnets.



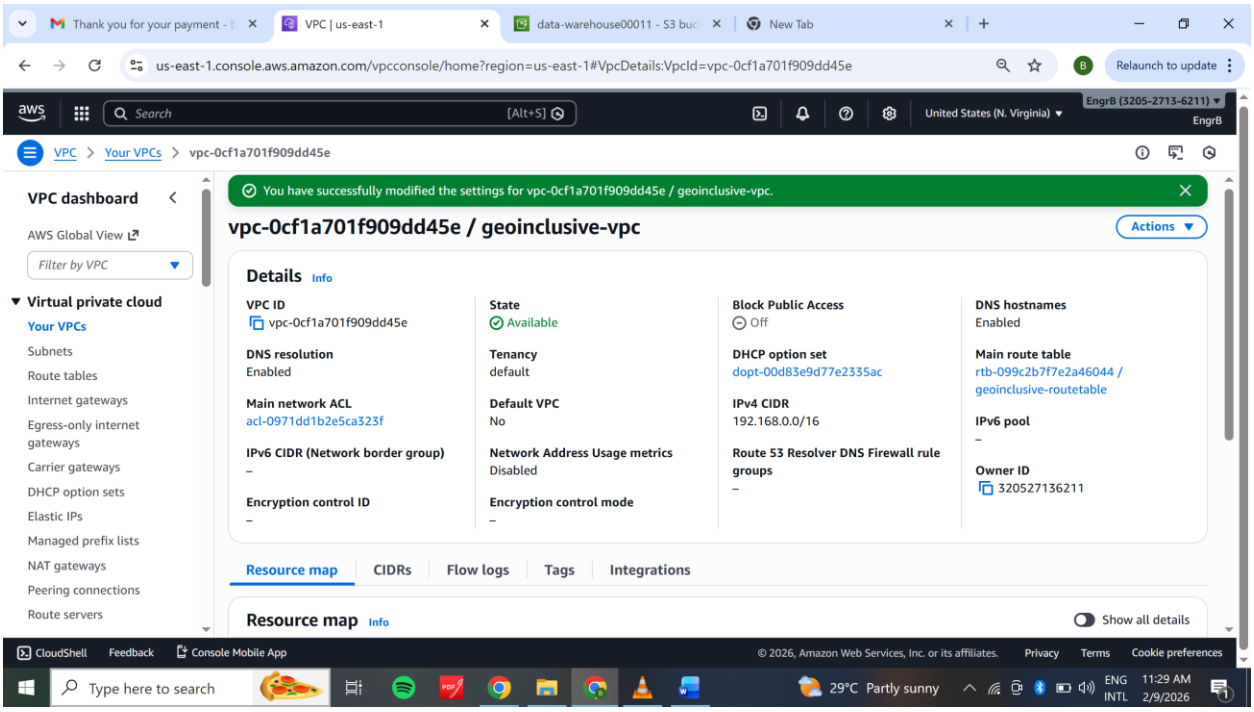
Step 17: To associate each subnet to the new NACLs, go to subnets, NACLs, edit subnet NACL associations, and replace them with the new NACLs created.



Step 18: Next, is to create security groups for each subnet. Go to security group - create security group, give it a name, and a description. For the inbound and outbound rules, allow everything and create. Give it a name tag and save. Go ahead and create SGs for all other subnets.

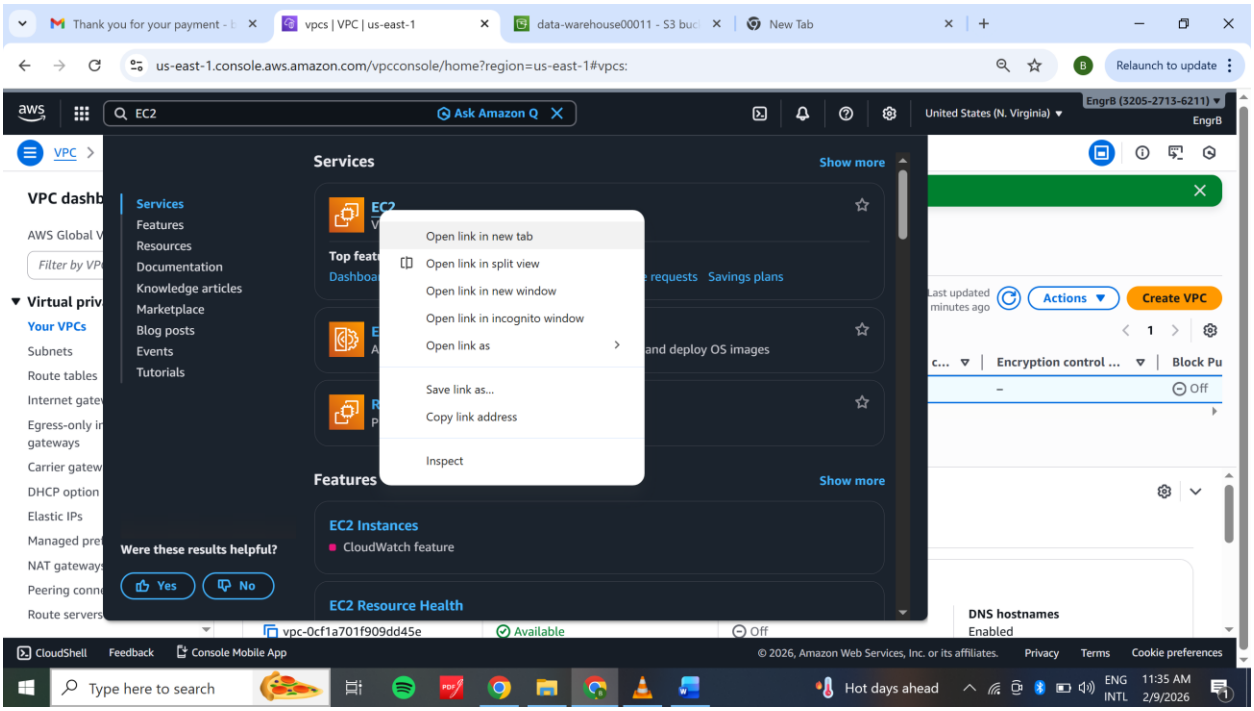


Step 19: Next step is to check the configurations of your VPC. Go to your VPC, click on it to check DNS, we have to enable it. Click on actions, edit VPC settings, and tick on both DNS properties and save. It will now be enabled.



DEPLOYING A VIRTUAL MACHINE IN THE VPC

Step 1: Search for EC2 instance and open it in another. Then click on launch instance.



Step 2: Put a name e.g windows machine, also choose the operating system e.g windows, choose the instance type (t3 micro) create a ssh key and call it windows key, it will be downloaded to your device. Save it.

For the network settings – edit - select VPC - select the subnets-select enable IP assign. For the security group, select exiting SG, click on the subnet to be used. For the advanced configuration, it will automatically assign one.

For storage, Click on Advance – Volume - Increase to your choice. The more, the cost (recommended GiB-GP3)

Click on launch instance

After the launch, ensure that all information is complete.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Relaunch to update

aws | Search [Alt+S] | United States (N. Virginia) | EngrB (3205-2713-6211) EngrB

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-044bbde7dfe13700e)

▸ Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

1 2 3 4 5 6

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Relaunch to update

aws | Search [Alt+S] | United States (N. Virginia) | EngrB (3205-2713-6211) EngrB

EC2 > Instances > Launch an instance

Name

Windows-machine

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Browse more AMIs
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Microsoft Windows Server 2025 Base
ami-06b5375e3af24939c (64-bit (x86))

Summary

Number of instances Info

1

Software Image (AMI)
Microsoft Windows Server 2025 ...read more
ami-06b5375e3af24939c

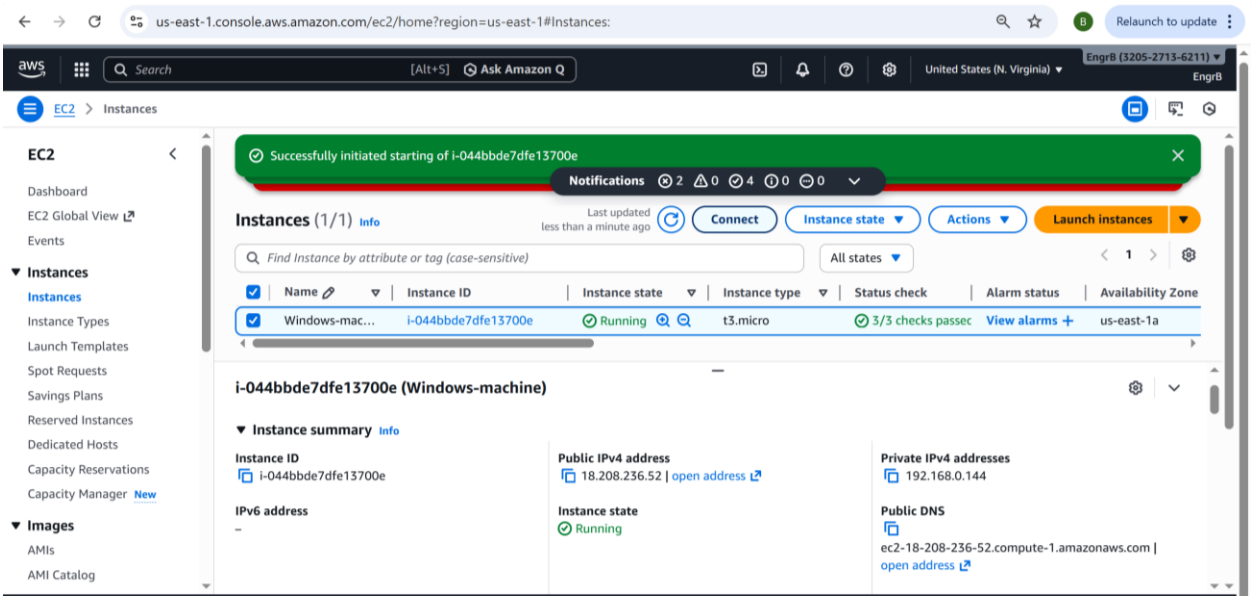
Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

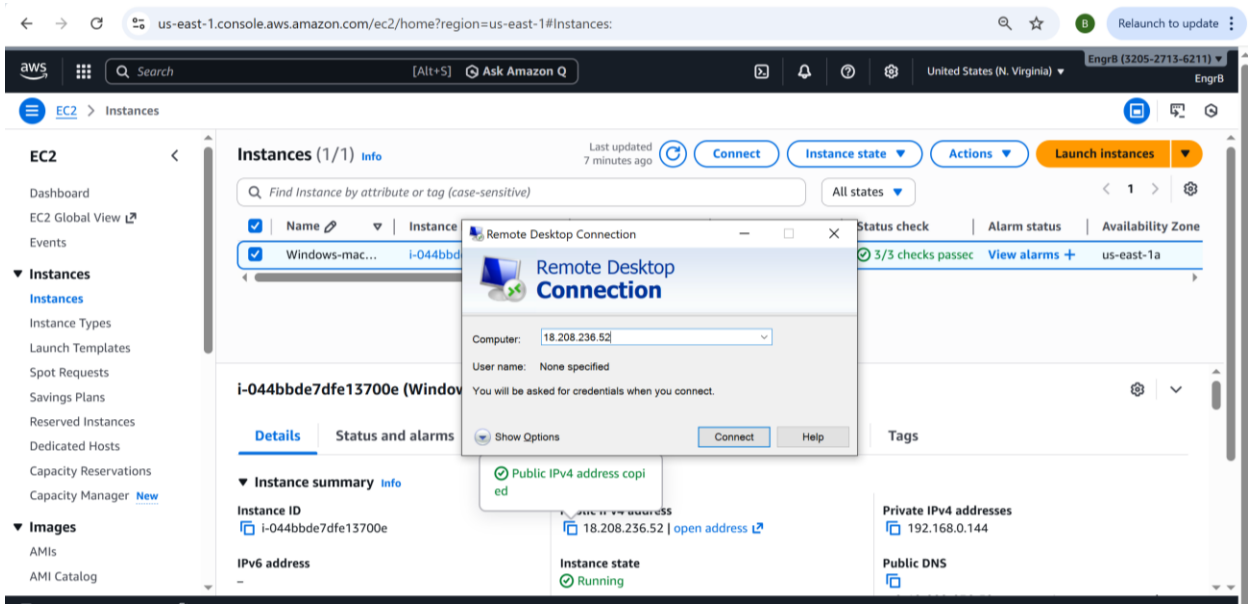
Storage (volumes)
1 volume(s) - 30 GiB

Cancel | Launch instance | Preview code

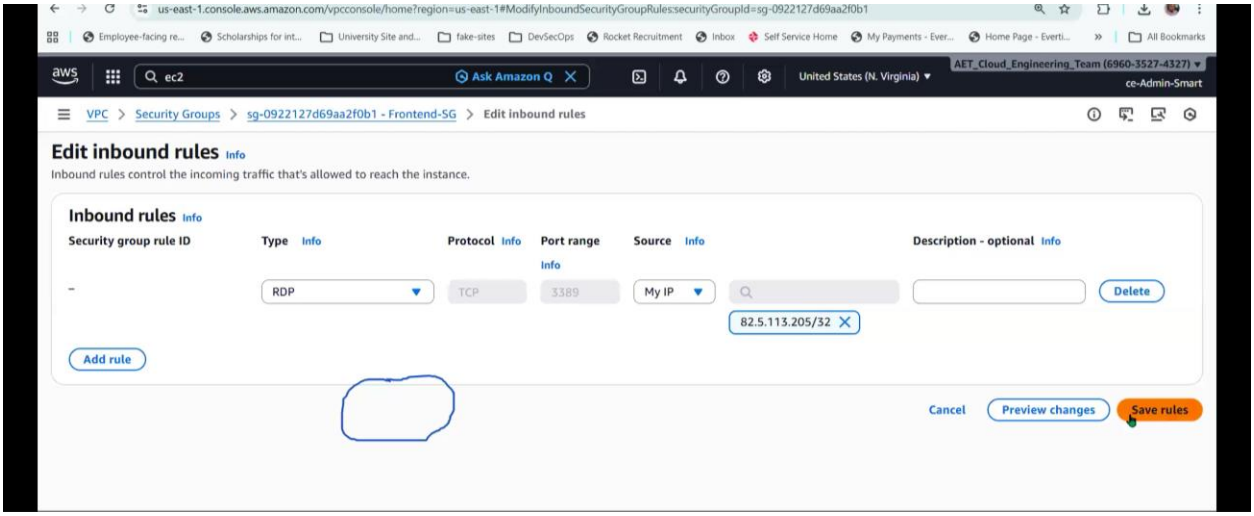
Step 3: The next step is to connect our machine. Click on connect on your instance.



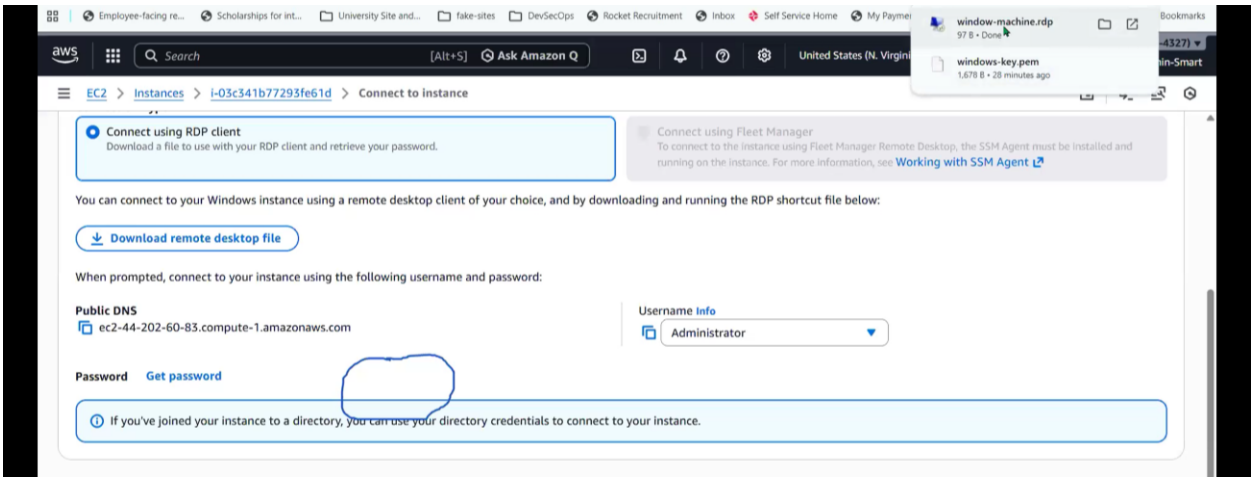
Step 4: Open the remote desktop connection in your local computer. Copy the public IP Address of the VM and then paste it in the dialogue box of the RDP. Then click on connect.

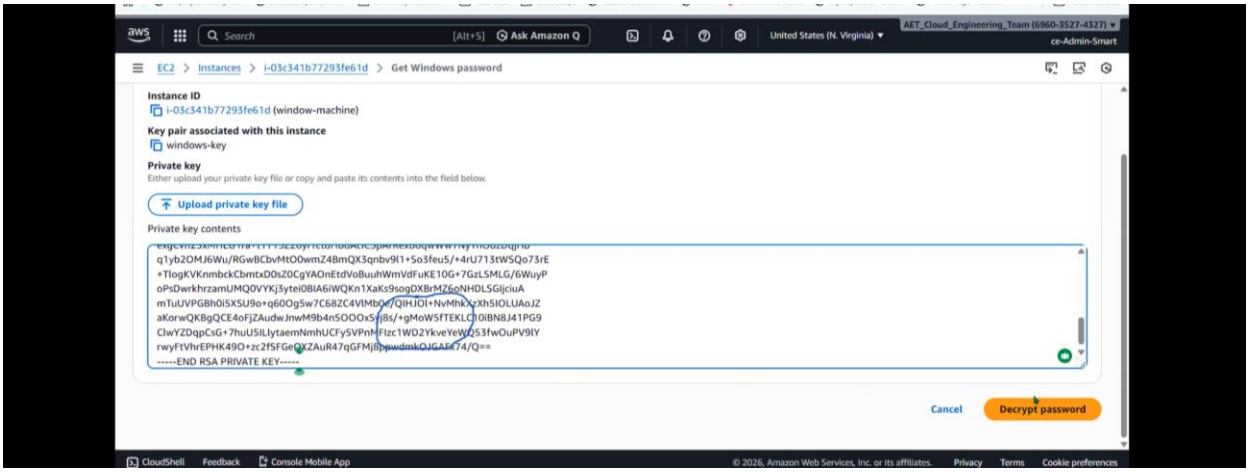


Step 5: For it to connect properly, go to NACLs. Go to flowlogs- reject logs - front-end-inbound rules - edit rules - add new rule and give it a number e.g 120, select RDP (3389) -allow - also get your public Ip address in the source and save. For outbound, allow all traffic. Also click on the front-end sg-inbound- edit -add RDP - source my IP address- save rules.

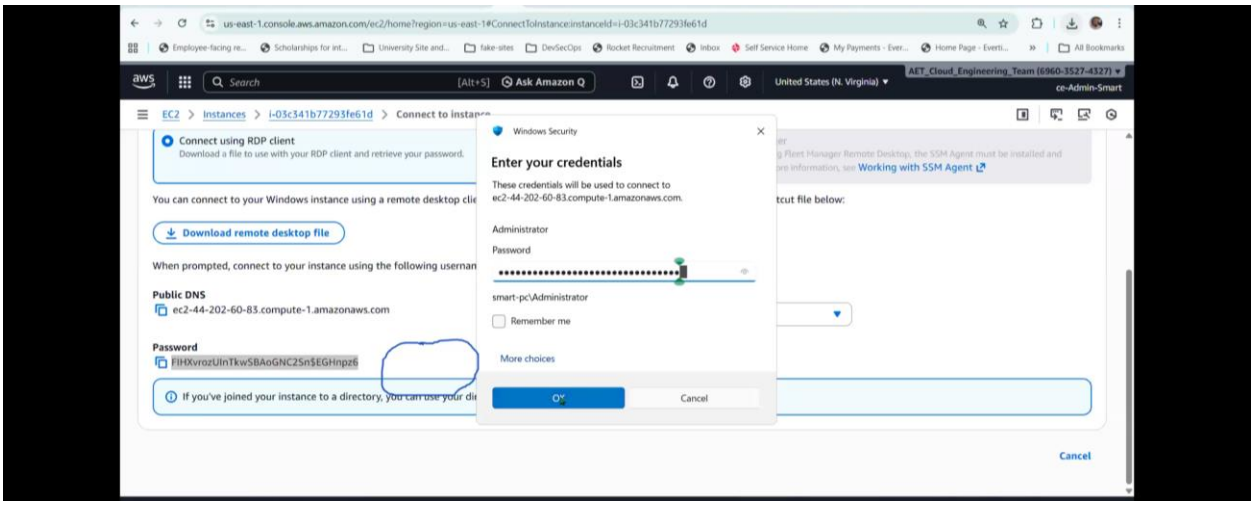


Step 6: When you connect now, it will connect but ask of a password. In the instance, go to Connect - RDP client, download the RD file. Then click on get Password, upload the downloaded RD, copy the keypair and paste, then click on Decrypt, a password will be given.

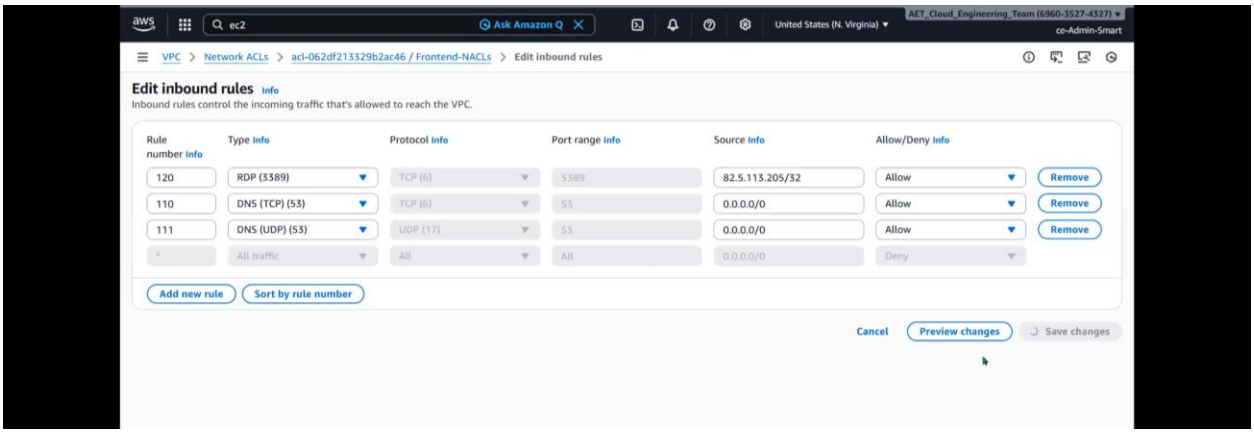




Step 7: Copy the password and paste it in your RDP dialogue box, and your virtual machine will be launched.



Step 8: To allow the virtual machine connect to the internet, Go to NACL, Allow DNS (53)in Inbound rules, create another rule and allow UDP DNS too. You can also check local resources in your virtual machine to enable audio and recording; you can also add a folder from your physical machine to the virtual.



Step 9: FINALLY, DELETE YOUR VIRTUAL MACHINE (Start with the Keypairs, delete them, Next to Volume, delete, Delete all NACLS, and all Security Groups).

Conclusion

Creating a VPC is a critical first step in building a secure and scalable cloud infrastructure in AWS. By following the steps outlined in this documentation, you now have a foundational network environment that can support various AWS services while maintaining isolation and control over traffic flow. A properly configured VPC enhances security, improves resource organization, and provides flexibility for future expansion. With your VPC in place, you are well-positioned to proceed with deploying compute, storage, and application services tailored to your workload requirements.